

Purit Hongjirakul

Location: Taipei City, Taiwan

[Website](#) | [LinkedIn](#) | [GitHub](#) | kyriosaaaph@gmail.com | (+886)989042217

EMBEDDED SYSTEMS ENGINEER

I am a Bachelor's graduate Embedded Systems Engineer with a strong foundation in electronics and microcontrollers. With a year of experience in **JavaScript and C/C++**, I specialize in blending software and hardware to create functional, efficient, and innovative machines. Currently, I am expanding my expertise into embedded IoT applications, connecting my projects to the internet and contributing to the ever-evolving technological landscape.

SKILLS

Hardware	: Microcontrollers (STM32, ESP32, Arduino), Embedded Systems, Circuit Design
Programming	: C, C++, Python, JavaScript, ReactJS
Tools	: Git, Linux, PyTorch, Visual Studio Code, NVIDIA Isaac Sim
Soft Skills	: Analytical Thinking, Oral Communication, Project Management
Languages	: English (Native), Thai (Native), Mandarin Chinese (Intermediate)

WORK EXPERIENCE

AP Computer Science Professor <i>Grace International School - Contract</i>	Mar 2025 – Present <i>Chiang Mai, Thailand</i>
- Taught Java & JavaScript to 11th and 12th graders. - Prepared them with complete and sufficient knowledge to pass CollegeBoard's AP Computer Science exam.	
Digital Twin R&D Engineer <i>Pegatron Corporation - Internship</i>	May 2024 – Oct 2024 <i>Taipei City, Taiwan</i>
- Developed a digital twin of a glue dispensing robot in NVIDIA Isaac Sim, enabling faster prototyping and reducing potential production errors. - Created a web application with REST APIs, JavaScript, and Python, including an AI chatbot with voice recognition.	

PROJECTS

Collision Prevention System <i>C/C++, Arduino & other electronics</i>	Apr 2025 <i>Personal</i>
- Created a compact, tactile system that helps prevent robotic vehicles from crashing into objects. Written in C/C++ and implements sensors such as ultrasonic, infrared, and photoresistor. - Soldered the Arduino board and other electrical components to a perfboard, ensuring the device stays durable while keeping its small form factor.	
PTCG Bot <i>Python</i>	Mar 2025 <i>Personal</i>
- Developed a news scraper bot using Python that finds articles relating to Pokemon to post on a chatroom. - Implemented different features such as a database, slash commands, and custom embedding to allow for the bot to be used by many communities without conflict. - This bot is currently being used in a community with over 31,000 members along with other smaller communities.	
Axis: A Gyroscope-Controlled RC Car <i>C++, ESP32 & other electronics</i>	Feb 2025 <i>Personal</i>
- Created a gyroscope-controlled RC car using ESP32 microcontrollers along with other electronic parts. - Created two devices: A gyroscope glove and a remote-controlled car. The gyroscope glove communicates with the car using Bluetooth, allowing for a wireless and seamless connection.	

EDUCATION

National Taipei University of Technology <i>Bachelor of Electrical Engineering and Computer Science – GPA 3.12</i>	Sep 2020 – Jul 2024 <i>Taipei City, Taiwan</i>
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